Some questions to think about…

- 1. The body speeds up and slows down the heart via GPCRs. Since both are GPCRs, how can they have such opposite effects?

- 2. Atenolol mentioned at the start of this lecture is used to prevent blood pressure rising too high. A raised blood pressure occurs when you have a ‘fight or flight’ response. So how might atenolol work?

Should I take one or two ibuprofen?



Qu. Should I take one or two ibuprofen?

Ans. This depends on the relationship between dose and response which itself depends on…

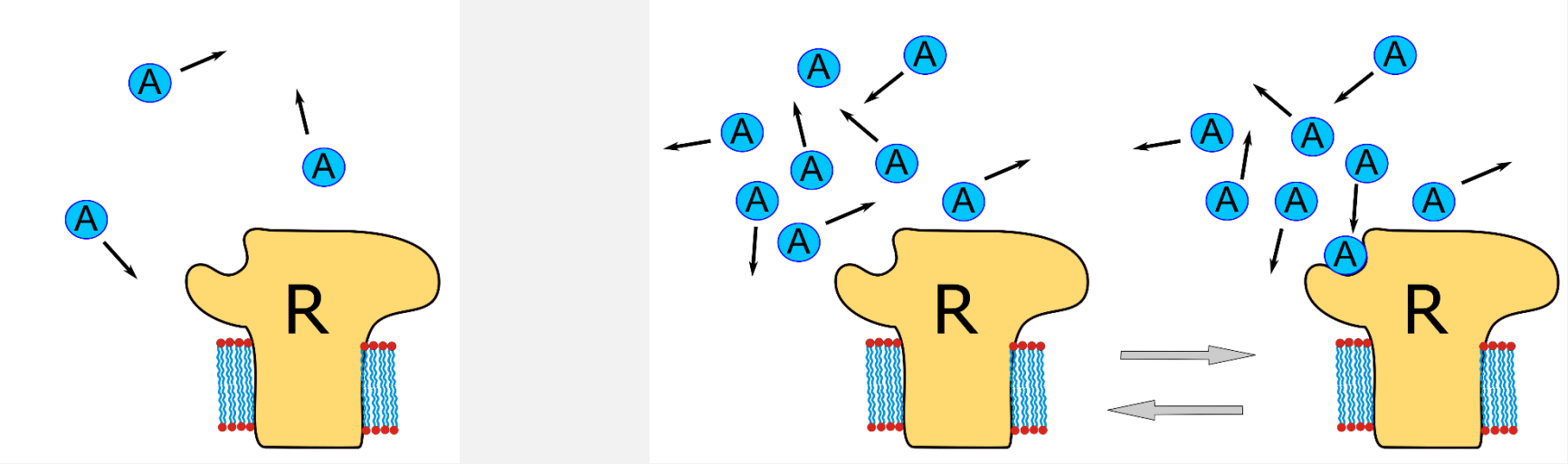
- Drug catabolism (how the drug is broken down by metabolic processes).
- The relationship between drug binding and tissue response.
- The concentration-dependence of the drug binding to its site of action.

The relationship between drug concentration and receptor occupancy…

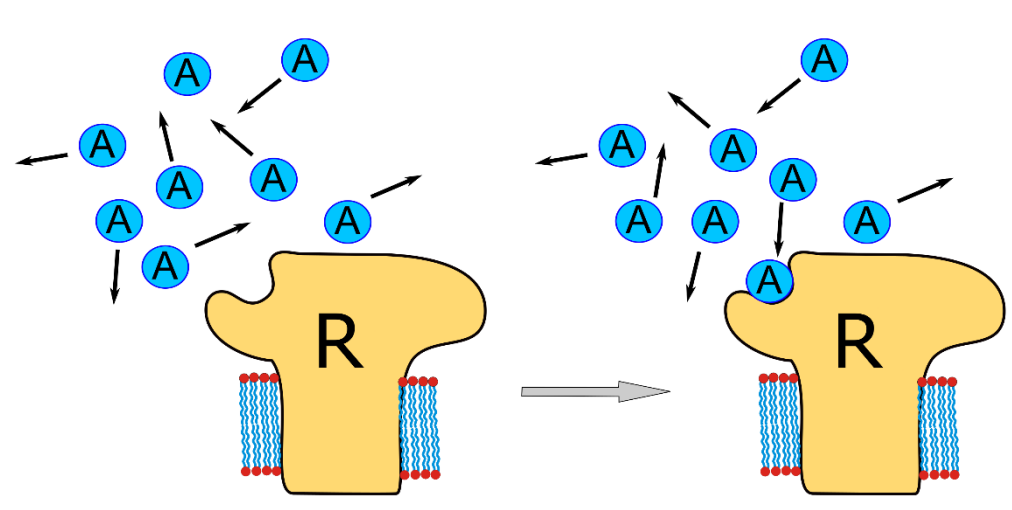
The law of mass action:

The rate of a reaction is proportional to the concentration of its reactants.

The law of mass action…



The relationship between drug concentration and receptor occupancy.



$$A + R \longrightarrow AR$$

The rate of forming a complex between A and R is $\propto$ [A][R]

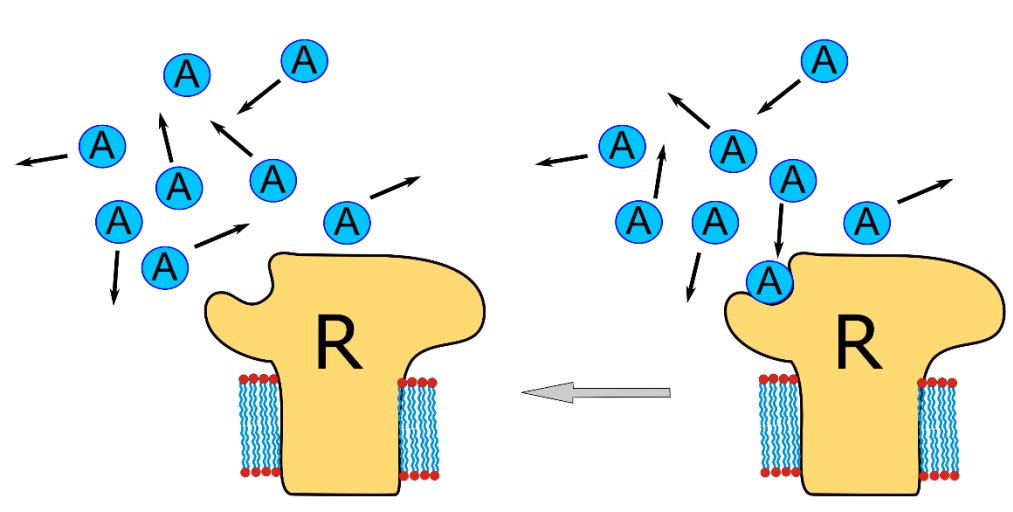
Getting rid of the '∝' sign...

The rate of forming a complex between A and R is $\propto$ [A][R]

The rate of forming a complex between A and R is = k_{+1}[A][R]

$$A + R \xrightarrow{k_{+1}} AR$$

Now to deal with the reverse reaction…



$$A + R \xleftarrow[k_{-1}]{} AR$$

The rate of breakdown of AR complex $\propto$ [AR]

The rate of breakdown of AR complex = k_{-1}[AR]

Naming things…

- We have named our rate constants k_{+1} and k_{-1}. Although this is in line with convention it is arbitrary (we could have chosen α and β for example). Whatever we call them:

- k_{+1} = the *association rate constant* (which has units of $M^{-1}\ s^{-1}$)

- k_{-1} = the *dissociation rate constant* (which has units of s^{-1}) respectively.

Let's consider our system at equilibrium…

$$A + R \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} AR$$

The rate of forming a complex between A and R is = rate of breakdown of AR

So… $$k_{+1}[A][R] = k_{-1}[AR]$$ Eq. 1

Remember, we are still trying to find the relationship between agonist concentration [A] and receptor occupancy. To do that we need another relationship…

We now consider the population of all receptors…

$$A + R \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} AR$$

$$[R]_T = [R] + [AR]$$

$$\frac{[R]_T}{[R]_T} = \frac{[R]}{[R]_T} + \frac{[AR]}{[R]_T}$$

$$1 \qquad p_R = \frac{[R]}{[R]_T} \qquad p_{AR} = \frac{[AR]}{[R]_T}$$

$$1 = p_R + p_{AR} \qquad \text{Eq. 2}$$

Returning to Equation 1 and dividing by $[R]_T$…

$$k_{+1}[A]\frac{[R]}{[R]_T} = k_{-1}\frac{[AR]}{[R]_T}$$

$$k_{+1}[A]p_R = k_{-1}p_{AR}$$

Now divide both sides by k_{+1} …

$$[A]p_R = K_A p_{AR}$$

where $K_A = \frac{k_{-1}}{k_{+1}}$

So…

$$p_R = \frac{K_A}{[A]}p_{AR}$$

Equation 3

$K_A = k_{-1}/k_{+1}$, is called the dissociation equilibrium constant and has units of concentration (e.g. M)

Combining equations 2 and three…

We had

$$1 = p_R + p_{AR}$$

But we know that

$$p_R = \frac{K_A}{[\mathrm{A}]} p_{AR}$$

So substituting in this relationship we have

$$1 = \frac{K_A}{[\mathrm{A}]} p_{AR} + p_{AR}$$

Equation 4

The Hill-Langmuir equation…

Factoring out p_{AR} gives:

$$1 = \left(\frac{K_A}{[A]} + 1\right) p_{AR}$$

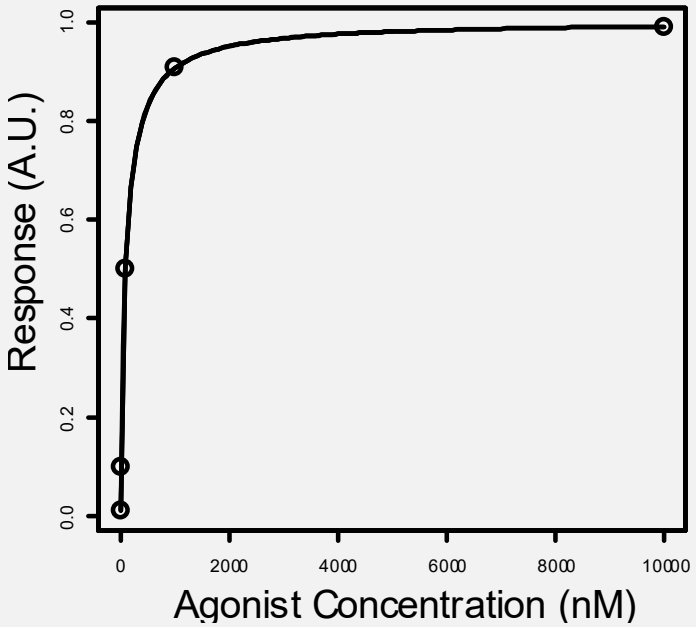
So…

$$p_{AR} = \frac{[A]}{[A] + K_A}$$ Equation 5

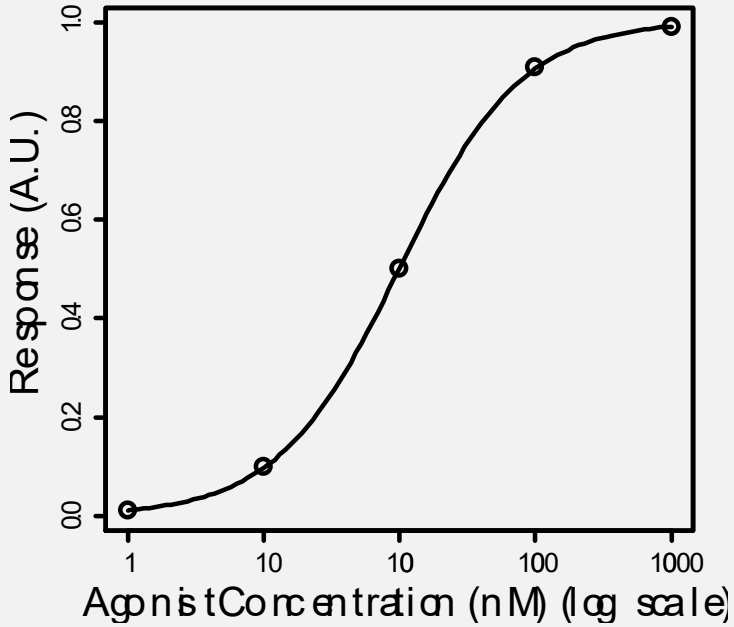
We did it! This is the relationship between drug concentration and receptor occupancy! It is called the Hill-Langmuir Equation.

What the Hill-Langmuir function looks like…

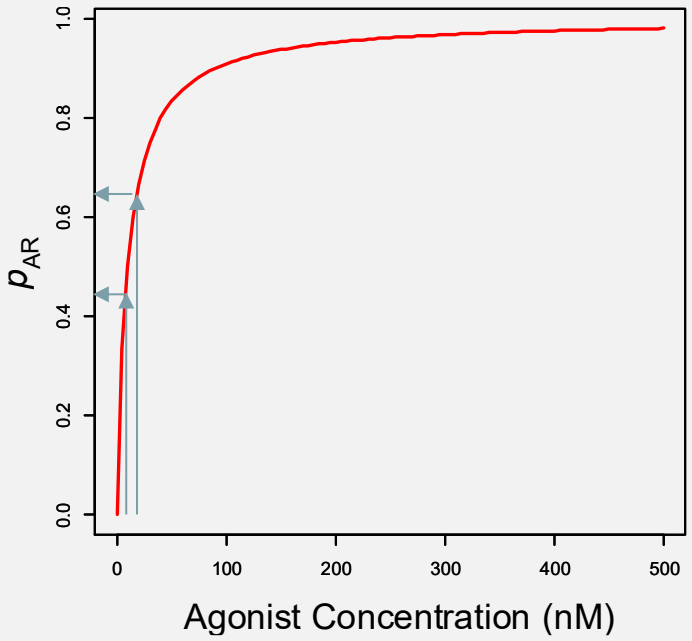
Linear scale

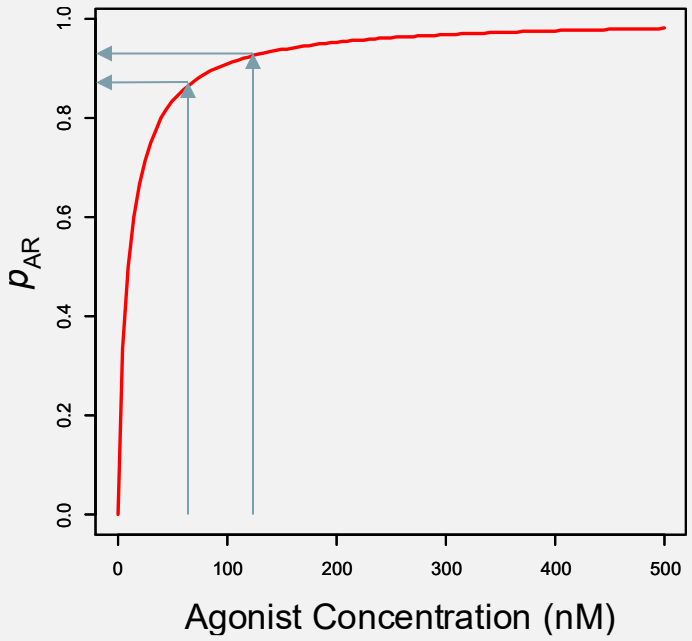


Log scale



So should we take one or two ibuprofen?…





What is the relationship between agonist occupancy of a receptor and the tissue response?

If we call the response of a tissue to a given agonist concentration '*y*', what happens if we assume:

$$\frac{y}{y_{max}} = p_{AR} \quad ?$$

If correct, then…

$$\frac{y}{y_{max}} = p_{AR} = \frac{[A]}{[A] + K_A}$$

Equation 6

Clarke and Hill had no modern computers for curve fitting…

$$\frac{y}{y_{max}} = p_{AR} = \frac{[\mathrm{A}]}{[\mathrm{A}] + K_A}$$

Multiply by y_{max} and ([A]+K_A)…

$$y([\mathrm{A}] + K_A) = y_{max}[\mathrm{A}]$$

Collecting the terms in [A]…

$$yK_A = y_{max}[\mathrm{A}] - y[\mathrm{A}]$$

So

$$yK_A = (y_{max} - y)[\mathrm{A}]$$

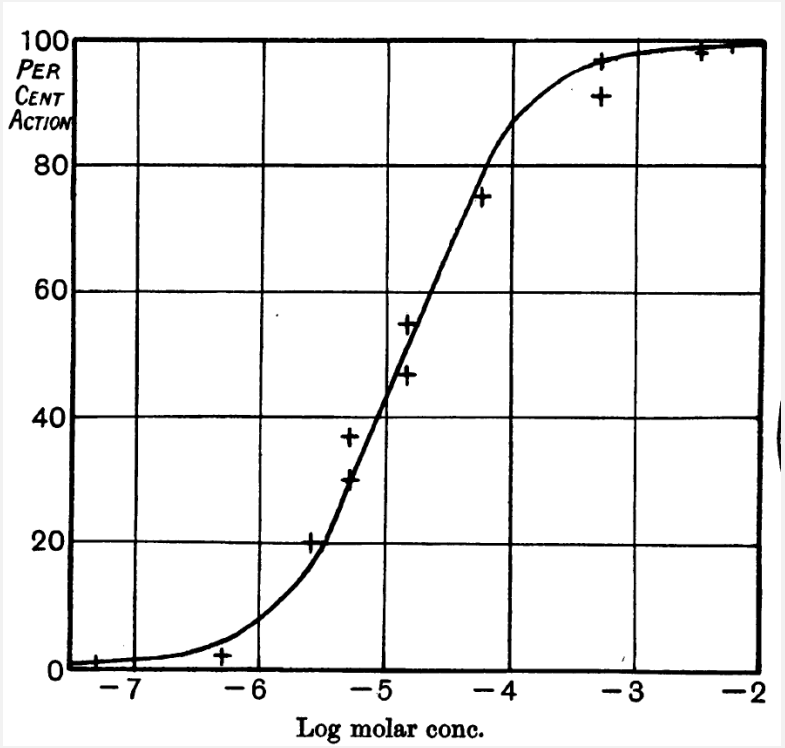
Clarke and Hill had no modern computers for curve fitting…

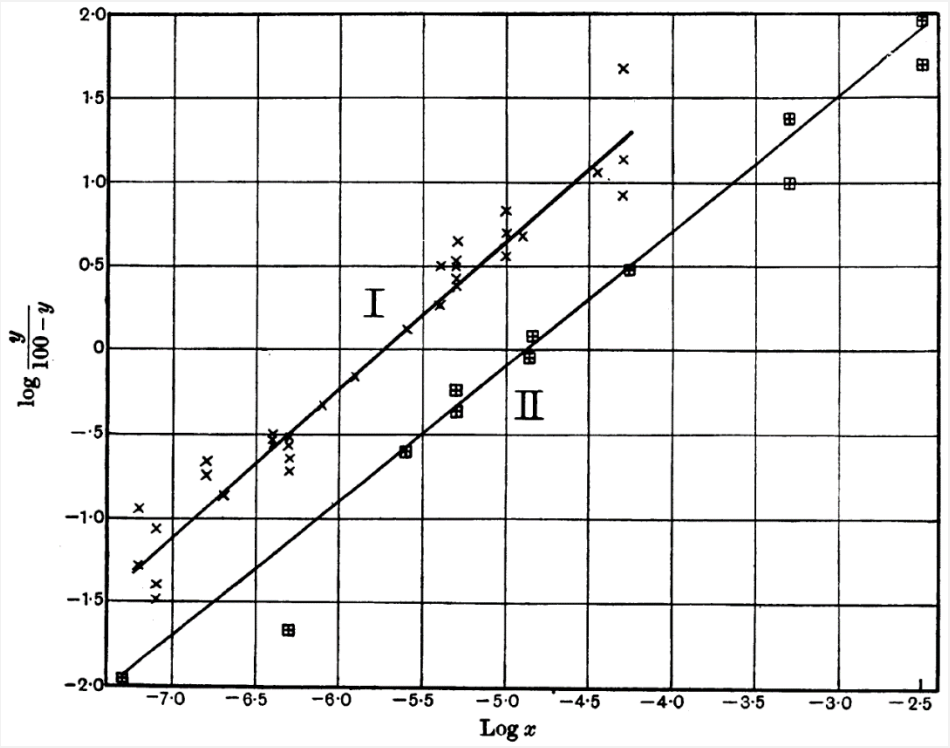
$$\frac{y}{y_{max} - y} = \frac{[\mathrm{A}]}{K_A}$$

Equation 7

$$\underbrace{log\left(\frac{y}{y_{max} - y}\right)}_{\mathrm{Y}} = \underbrace{\log[A] - log\ K_A}_{\mathrm{X}}$$

Here's what they found…





A straight line but not with unity slope….

$$\underbrace{log\left(\frac{y}{y_{max}-y}\right)}_{Y} = nH * \underbrace{\log[A]}_{X} - log\ C$$

The slope = Hill coefficient

A constant representing the intercept

Manipulating the logs…

$$log\left(\frac{y}{y_{max}-y}\right) = \log[\mathrm{A}]^{nH} - log\ C$$

And…

$$log\left(\frac{y}{y_{max}-y}\right) = \log\frac{[\mathrm{A}]^{nH}}{C}$$

A straight line but not with unity slope….

$$log\left(\frac{y}{y_{max}-y}\right) = \log\frac{[\mathrm{A}]^{nH}}{C}$$

So

$$\left(\frac{y}{y_{max}-y}\right) = \frac{[\mathrm{A}]^{nH}}{C}$$

Now multiply both sides by $C^*(y_{max}- y)$…

$$yC = (y_{max} - y)[\mathrm{A}]^{nH}$$

Collecting terms in gives y…

$$y([\mathrm{A}]^{nH} + C) = y_{max}[\mathrm{A}]^{nH}$$

A straight line but not with unity slope….

$$y([\mathrm{A}]^{nH} + C) = y_{max}[\mathrm{A}]^{nH}$$

Now divide by $([\mathrm{A}]^{nH} + C)$ and we get:

$$\frac{y}{y_{max}} = \frac{[\mathrm{A}]^{nH}}{([\mathrm{A}]^{nH} + C)}$$

We can see at once that $[\mathrm{A}]^{nH} = C$ then $y/y_{max} = C/(C+C) = 0.5$, the half-maximal response. Hence C can be re-written as $[\mathrm{A}_{50}]^{nH}$ giving

$$\frac{y}{y_{max}} = \frac{[\mathrm{A}]^{nH}}{([\mathrm{A}]^{nH} + [\mathrm{A}_{50}]^{nH})}$$

The Hill Equation

A straight line but not with unity slope….

$$\frac{y}{y_{max}} = \frac{[\mathrm{A}]^{nH}}{([\mathrm{A}]^{nH} + [\mathrm{A}_{50}]^{nH})}$$

$$\underbrace{log\left(\frac{y}{y_{max} - y}\right)}_{\text{Y}} = nH * \underbrace{\log[A]}_{\text{X}} - nH * log\ [A]_{50}$$

Hill coefficient

The agonist concentration required for half-maximal response

Some questions to think about…

- 1. If a drug were to bind irreversibly, could you measure the affinity (K_B value)?

- 2. Why might the Hill coefficient be different from one?